Jun-Kun Chen

Ph.D. Candidate in Computer Science | Expected Dec. 2026

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Legal name: Junkun Chen | Publications: Jun-Kun Chen / J.-K. Chen

Research Focus

Research in **diffusion-based controllable generation**: supervision and training design for learning new generative capabilities, including visual control and temporally consistent synthesis. Current work connects language reasoning with **visual world models** through multimodal post-training.

Education

University of Illinois Urbana-Champaign — **2021 – Dec. 2026 (expected)**
Ph.D. Candidate in Computer Science; Advisor: Prof. **Yu-Xiong Wang** — Urbana, IL
Master of Computer Science, Dec. 2023 (earned during Ph.D. study)

Tsinghua University — **2017 – 2021**
B.Eng. in Computer Science, Yao Class (IIIS); Major GPA: 3.94/4.0 — Beijing, China

Selected Publications

1. Virtual Fitting Room: Generating Arbitrarily Long Videos of Virtual Try-On from a Single Image ↗ — **NeurIPS 2025**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Sep. 2025)

2. Dress&Dance: Dress up and Dance as You Like It ↗ — **WACV 2027**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Aug. 2025)

3. ProEdit: Simple Progression is All You Need for High-Quality 3D Scene Editing ↗ — **NeurIPS 2024**
Jun-Kun Chen, Yu-Xiong Wang — Project | arXiv (Nov. 2024)

4. ConsistDreamer: 3D-Consistent 2D Diffusion for High-Fidelity Scene Editing ↗ — **CVPR 2024**
Jun-Kun Chen, Samuel Rota Bulò, Norman Müller, Lorenzo Porzi, Peter Kontschieder, Yu-Xiong Wang — Project | arXiv (Jun. 2024)

5. Instruct 4D-to-4D: Editing 4D Scenes as Pseudo-3D Scenes Using 2D Diffusion ↗ — **CVPR 2024**
Linzhan Mou*, **Jun-Kun Chen***, Yu-Xiong Wang — Project | arXiv (Jun. 2024)

6. HDEdit: Editing Videos and 3D Scenes with Video Diffusion Through Hierarchical Task Decomposition ↗ — **3DV 2026**
Yanming Zhang*, **Jun-Kun Chen***, Jipeng Lyu, Yu-Xiong Wang — Project | Earlier version (arXiv, Mar. 2025)

*Equal contribution. Earlier publications also appear as Junkun Chen / J. Chen.

Industry Experience

Meta Reality Labs — **May–Dec. 2025; May–Aug. 2023**
Research Scientist Intern (2025 continued part-time Aug.–Dec.) — Bay Area, CA, U.S. / Zurich, Switzerland

- **2025:** Designed a **GPU-role-separated pipeline** for data processing and model training at **128–256-GPU scale**, supporting 3D/video-generation research on Meta's distributed infrastructure.
- **2023:** Designed and implemented **self-supervised diffusion adaptation** and an asynchronous training pipeline for *ConsistDreamer* (CVPR 2024).

SpreeAI — **May 2024 – May 2025**
AI Research Intern — Remote, U.S.

- **Dress&Dance**: designed **target-steered supervision and staged training** for garment, identity, and motion control in video diffusion, generating 5-second try-on videos at **1152×720, 24 FPS**.
- **Virtual Fitting Room** (NeurIPS 2025): designed **reference-conditioned training from short-clip pairs**, enabling anchored autoregression for identity-consistent video up to **90 seconds**.

Current Research: Generative World Models

Research direction: connecting language reasoning with visual generation, using predicted outcomes as training feedback.

Computer Use — Action-Conditioned Image-Diffusion World Model

- Built a next-screen world model on SenseNova MoT: a language Reasoner describes the action's effects, and a Generator adapted by **diffusion post-training** renders the next screen.
- Formulated **teacher-grounded supervision** for action effects and content preservation, now used to train the Reasoner.

Physics — Video-Diffusion World Model for Physical Events

- Adapted **Cosmos3 video diffusion** to predict physical events from past RGB frames and structured event descriptions.
- Implemented **teacher distillation** from a full-video teacher to a Reasoner that sees only past frames.

Robotics — Motion-Conditioned Video Diffusion World Model

- Trained a **motion-conditioned video world model** on simulated and real robot data (MimicGen, RoboTwin, DROID, RH20T, AgiBot) to predict interaction video from a scene, recent frames, and rendered robot motion.
- Implemented **visual-proxy and paired-correspondence conditioning**; studying adaptation to unseen robot setups.

Technical Skills & Honors

Tools & Methods: PyTorch; Python; C++; FSDP2; Slurm; image/video diffusion (DiT, rectified flow); LoRA; VLMs; diffusion post-training (SFT, DiffusionNFT); teacher distillation; human/VLM-judge evaluation; paired-bootstrap statistics.
Competitions: ICPC EC-Final Gold (4th); ICPC Qingdao Gold (2nd); CCPC Harbin Gold; NOI Gold; NOI/ICPC problem setter.

Selected Research Contributions

Learning Visual Control and Long-Horizon Generation — Dress&Dance and VFR

- **Dress&Dance:** designed **target-steered supervision and staged curricula** for new visual controls, achieving **20–59% lower FVD** than a Kling 1.6 try-on/animation pipeline on Internet and captured benchmarks (Table C.4).
- Designed multimodal conditioning and **synthetic-condition/real-target triplets**; implemented hybrid image/video training and progressive schedules to add garment and identity control while retaining pretrained priors.
- **Virtual Fitting Room:** designed **short-clip pair sampling** to learn reference-conditioned appearance and prefix continuation, enabling generated 360° anchors at inference for identity-consistent video up to **90 seconds**.

General-Purpose Diffusion Editing — HDEdit and ProEdit

- **HDEdit:** designed **hierarchical task decomposition** to separate instruction following from content preservation, combining LLM scheduling with interpolation refinement for training-free video and scene editing.
- **ProEdit:** designed and implemented tuning-free preservation control and a **progressive editing workflow** with previews and edit-strength control for high-fidelity geometry and appearance editing.

Multi-View and Temporal Consistency — ConsistDreamer and Instruct 4D-to-4D

- **ConsistDreamer:** designed **cross-view self-supervision and staged diffusion adaptation** using warped-view targets, geometry-aware ControlNet/LoRA, and consistency losses across the **denoising trajectory**.
- Implemented an **asynchronous multi-GPU pipeline** that overlaps synchronous diffusion training and image generation with buffered scene fitting on a dedicated GPU.
- **Instruct 4D-to-4D:** extended 2D diffusion to long-term dynamic scenes through **anchor-aware attention** and optical-flow-guided sliding windows.

Geometric Visual Representations — NeuralEditor and PointTree

- Built point-based radiance-field and relaxed k-d-tree representations for editable scene geometry and **transformation-robust point-cloud encoding**.

Additional Publications

7. SceneCraft: Layout-Guided 3D Scene Generation ↗ **NeurIPS 2024**
Xiuyu Yang*, Yunze Man*, **Jun-Kun Chen**, Yu-Xiong Wang Project | arXiv (Oct. 2024)

8. Contrastive Learning Relies More on Spatial Inductive Bias Than Supervised Learning: An Empirical Study ↗ **ICCV 2023**
Yuanyi Zhong*, Haoran Tang*, **Jun-Kun Chen***, Yu-Xiong Wang

9. NeuralEditor: Editing Neural Radiance Fields via Manipulating Point Clouds ↗ **CVPR 2023**
Jun-Kun Chen*, Jipeng Lyu*, Yu-Xiong Wang

10. PointTree: Transformation-Robust Point Cloud Encoder with Relaxed K-D Trees ↗ **ECCV 2022**
Jun-Kun Chen, Yu-Xiong Wang

11. Is Self-Supervised Contrastive Learning More Robust Than Supervised Learning? ↗ **ICML 2022 Workshop**
Yuanyi Zhong*, Haoran Tang*, **Jun-Kun Chen**, Jian Peng, Yu-Xiong Wang

12. TorchDrug: A Powerful and Flexible Machine Learning Platform for Drug Discovery ↗ **arXiv preprint**, 2022
Zhaocheng Zhu, Chence Shi, Zuobai Zhang, ..., **Jun-Kun Chen**, ..., Jian Tang

13. RNNLogic: Learning Logic Rules for Reasoning on Knowledge Graphs ↗ **ICLR 2021**
Meng Qu*, **Jun-Kun Chen***, Louis-Pascal Xhonneux, Yoshua Bengio, Jian Tang

14. Pre-training Assisted Scene-aware Dialogue Generation ↗ **DSTC8 Workshop at AAAI 2020**
Jun-Kun Chen, Huang He, Siqi Bao, Fan Wang

Earlier Experience

Mila – Quebec AI Institute **Feb. – Aug. 2020**
Research Intern; knowledge-graph reasoning with Prof. Jian Tang Montreal, Canada

Baidu, Natural Language Processing Department **July 2019 – Jan. 2020**
Research / Development Intern; scene-aware dialogue generation (DSTC8 top-3 system) Beijing, China

Competitions, Leadership & Coursework

Programming Competitions: ICPC EC-Final 2018, Gold Medal (4th); ICPC Qingdao Regional 2018, Gold Medal (2nd); CCPC Harbin Regional 2017, Gold Medal; NOI 2016, Gold Medal.
Additional Distinctions: CTSC 2017 (7th); CTSC 2016 Gold (8th); NOI Winter Camp 2016 Gold (3rd).
Leadership & Problem Setting: Set 9 contest problems (2018–2019), including hardest-tier tasks at NOI 2018 and NOI 2019, plus problems for CTSC, the NOI Winter Camp, the Tsinghua University Programming Contest (THUPC), ICPC Hong Kong, and CCPC Qinhuangdao. Vice Chair, Tsinghua Student Association of Algorithms and Competitions (2018–2019).
Selected Coursework (Generative Models): Deep Generative Models (A+); Generative Models for Biomedicine and Life Sciences (A).
Vision & 3D: 3D Vision (A+); Deep Learning for Computer Vision (A); Computer Vision (A); Computational Photography (A+).
Learning: Meta-Learning (A+); Machine Learning (A).